

# Homework Chapter 4

## 1 Problem 1

### 1.1 Question 1

The nearest-neighbor tight-binding Hamiltonian in real space is

$$H = t \sum_{\mathbf{R}} \sum_{j=1}^3 (|A, \mathbf{R}\rangle \langle B, \mathbf{R} + \boldsymbol{\delta}_j| + |B, \mathbf{R} + \boldsymbol{\delta}_j\rangle \langle A, \mathbf{R}|),$$

where  $t$  is the nearest-neighbor hopping parameter and  $\boldsymbol{\delta}_j$  are the three vectors connecting an  $A$  site to its three nearest-neighbor  $B$  sites.

Using the Bloch basis,

$$|A, \mathbf{k}\rangle = \frac{1}{\sqrt{N}} \sum_{\mathbf{R}} e^{i\mathbf{k}\cdot\mathbf{R}} |A, \mathbf{R}\rangle,$$

and

$$|B, \mathbf{k}\rangle = \frac{1}{\sqrt{N}} \sum_{\mathbf{R}} e^{i\mathbf{k}\cdot\mathbf{R}} |B, \mathbf{R}\rangle,$$

the diagonal matrix elements are

$$H_{AA} = H_{BB} = 0,$$

because the nearest-neighbor model contains no  $A - A$  or  $B - B$  hopping.

The off-diagonal matrix element is

$$H_{AB} = \langle A, \mathbf{k} | H | B, \mathbf{k} \rangle = t \sum_{j=1}^3 e^{i\mathbf{k}\cdot\boldsymbol{\delta}_j}.$$

Defining

$$f(\mathbf{k}) = \sum_{j=1}^3 e^{i\mathbf{k}\cdot\boldsymbol{\delta}_j},$$

we obtain

$$H_{AB} = t f(\mathbf{k}).$$

Since the Hamiltonian is Hermitian,

$$H_{BA} = H_{AB}^* = tf^*(\mathbf{k}).$$

Therefore, the Bloch Hamiltonian for a single graphene layer is

$$H_{1L}(\mathbf{k}) = \begin{pmatrix} 0 & tf(\mathbf{k}) \\ tf^*(\mathbf{k}) & 0 \end{pmatrix}.$$

The structure factor can be written as

$$f(\mathbf{k}) = e^{i\mathbf{k}\cdot\boldsymbol{\delta}_1} + e^{i\mathbf{k}\cdot\boldsymbol{\delta}_2} + e^{i\mathbf{k}\cdot\boldsymbol{\delta}_3}.$$

The three terms represents the three nearest-neighbor  $p_z$ -orbital interactions between one  $A$  carbon atom and its three neighboring  $B$  carbon atoms, located at displacements  $\boldsymbol{\delta}_1$ ,  $\boldsymbol{\delta}_2$ , and  $\boldsymbol{\delta}_3$ .

## 1.2 Question 2

To find the energy eigenvalues, we diagonalize the single-layer graphene Hamiltonian

$$H_{1L}(\mathbf{k}) = \begin{pmatrix} 0 & tf(\mathbf{k}) \\ tf^*(\mathbf{k}) & 0 \end{pmatrix}.$$

The eigenvalues satisfy the equation

$$\det(H_{1L}(\mathbf{k}) - EI) = 0.$$

Therefore,

$$\det \begin{pmatrix} -E & tf(\mathbf{k}) \\ tf^*(\mathbf{k}) & -E \end{pmatrix} = 0.$$

Evaluating the determinant gives

$$E^2 - t^2 f(\mathbf{k})f^*(\mathbf{k}) = 0.$$

Using

$$f(\mathbf{k})f^*(\mathbf{k}) = |f(\mathbf{k})|^2,$$

we obtain

$$E^2 = t^2 |f(\mathbf{k})|^2.$$

Taking the square root gives the two energy eigenvalues

$$E_{\pm}(\mathbf{k}) = \pm |t| |f(\mathbf{k})|.$$

and,

$$E_{\pm}(\mathbf{k}) = \pm |t f(\mathbf{k})|.$$

These two eigenvalues correspond to the lower and upper energy bands of a single graphene layer.

To verify the analytical result numerically, we choose

$$\mathbf{k} = (0.30, 0.40) \text{ \AA}^{-1}.$$

Using the analytical expression

$$E_{\pm}(\mathbf{k}) = \pm |t| |f(\mathbf{k})|,$$

we obtain

$$E_- = -7.11068512 \text{ eV}, \quad E_+ = 7.11068512 \text{ eV}.$$

The same Hamiltonian was then diagonalized numerically using `numpy.linalg.eigvalsh`, which gives

$$E_- = -7.11068512 \text{ eV}, \quad E_+ = 7.11068512 \text{ eV}.$$

The numerical eigenvalues therefore agree with the analytical result.

### 1.3 Question 3

Using the result from Question 2,

$$E_{\pm}(\mathbf{k}) = \pm |t| |f(\mathbf{k})|,$$

we evaluate the structure factor and the corresponding energies at the high-symmetry points  $\Gamma$ ,  $K$ , and  $M$ .

For the  $\Gamma$  point,

$$\Gamma = (0, 0),$$

the structure factor is

$$f(\Gamma) = 3,$$

so

$$|f(\Gamma)| = 3.$$

Therefore,

$$E_-(\Gamma) = -8.10 \text{ eV}, \quad E_+(\Gamma) = 8.10 \text{ eV}.$$

For the  $K$  point,

$$K = \left( \frac{4\pi}{3\sqrt{3}d}, 0 \right),$$

the numerical evaluation gives

$$f(K) \approx 3.89 \times 10^{-16},$$

which is zero within numerical precision. Therefore,

$$|f(K)| \approx 0,$$

and

$$E_-(K) \approx 0, \quad E_+(K) \approx 0.$$

For the  $M$  point,

$$M = \left( \frac{\pi}{\sqrt{3}d}, \frac{\pi}{3d} \right),$$

the structure factor is

$$f(M) = 0.5 + 0.866025 i,$$

so

$$|f(M)| = 1.$$

Therefore,

$$E_-(M) = -2.70 \text{ eV}, \quad E_+(M) = 2.70 \text{ eV}.$$

Thus, the values at the three high-symmetry points are

$$|f(\Gamma)| = 3, \quad |f(K)| = 0, \quad |f(M)| = 1,$$

with corresponding energies

$$E_{\pm}(\Gamma) = \pm 8.10 \text{ eV},$$

$$E_{\pm}(K) = 0,$$

and

$$E_{\pm}(M) = \pm 2.70 \text{ eV}.$$

## 1.4 Question 4

The two graphene bands were plotted along the high-symmetry path

$$\Gamma \rightarrow K \rightarrow M \rightarrow \Gamma$$

using

$$E_{\pm}(\mathbf{k}) = \pm |t| |f(\mathbf{k})|.$$

At the  $\Gamma$  point, the two bands are separated by the largest energy in this path,

$$E_{\pm}(\Gamma) = \pm 8.10 \text{ eV}.$$

As the path approaches  $K$ , the two bands move toward each other and meet at

$$E_+(K) = E_-(K) = 0.$$

The bands then separate again between  $K$  and  $M$ , with

$$E_{\pm}(M) = \pm 2.70 \text{ eV},$$

and return to

$$E_{\pm}(\Gamma) = \pm 8.10 \text{ eV}.$$

The second panel shows the positive-energy band  $E_+(\mathbf{k})$  over a two-dimensional region of reciprocal space containing both  $K$  and  $K'$ .

The positive band reaches zero energy at both points,

$$E_+(K) = E_+(K') = 0.$$

Therefore, the conduction and valence bands touch at the two inequivalent points  $K$  and  $K'$ . These points are the Dirac points of the nearest-neighbor graphene model.

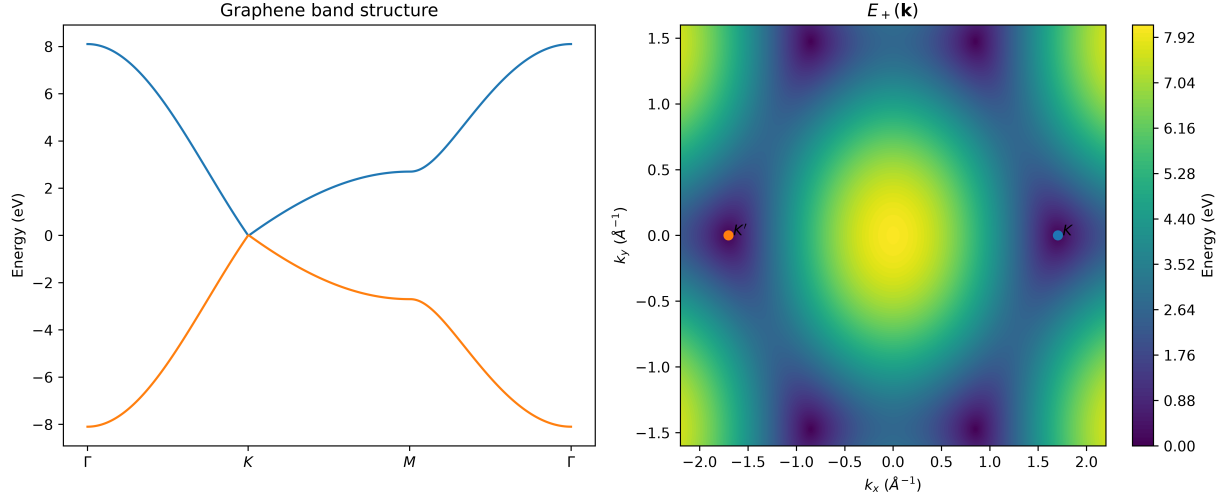


Figure 1: Graphene band structure along  $\Gamma \rightarrow K \rightarrow M \rightarrow \Gamma$  and the positive-energy band  $E_+(\mathbf{k})$  in a two-dimensional region containing the Dirac points  $K$  and  $K'$ .

## 1.5 Question 5

Near the Dirac point  $K$ , write

$$\mathbf{k} = \mathbf{K} + \mathbf{q},$$

where  $\mathbf{q}$  is small. The structure factor is

$$f(\mathbf{k}) = \sum_{j=1}^3 e^{i\mathbf{k} \cdot \boldsymbol{\delta}_j}.$$

Thus,

$$f(\mathbf{K} + \mathbf{q}) = \sum_{j=1}^3 e^{i\mathbf{K} \cdot \boldsymbol{\delta}_j} e^{i\mathbf{q} \cdot \boldsymbol{\delta}_j}.$$

For small  $\mathbf{q}$ ,

$$e^{i\mathbf{q} \cdot \boldsymbol{\delta}_j} \approx 1 + i\mathbf{q} \cdot \boldsymbol{\delta}_j.$$

Therefore,

$$f(\mathbf{K} + \mathbf{q}) \approx \sum_{j=1}^3 e^{i\mathbf{K} \cdot \boldsymbol{\delta}_j} + i \sum_{j=1}^3 e^{i\mathbf{K} \cdot \boldsymbol{\delta}_j} (\mathbf{q} \cdot \boldsymbol{\delta}_j).$$

Since

$$f(\mathbf{K}) = 0,$$

the first term vanishes. Using the nearest-neighbor vectors gives

$$f(\mathbf{K} + \mathbf{q}) \approx -\frac{3d}{2}(q_x - iq_y).$$

Hence,

$$|f(\mathbf{K} + \mathbf{q})| \approx \frac{3d}{2} \sqrt{q_x^2 + q_y^2} = \frac{3d}{2} |\mathbf{q}|.$$

Using

$$E_{\pm}(\mathbf{k}) = \pm |t| |f(\mathbf{k})|,$$

the dispersion near  $K$  becomes

$$E_{\pm}(\mathbf{K} + \mathbf{q}) \approx \pm \frac{3}{2} |t| d |\mathbf{q}|.$$

Comparing this with

$$E_{\pm} = \pm \hbar v_F |\mathbf{q}|,$$

we obtain

$$\hbar v_F = \frac{3}{2} |t| d.$$

Therefore,

$$v_F = \frac{3|t|d}{2\hbar}.$$

Using

$$d = \frac{a}{\sqrt{3}} = \frac{2.46 \text{ \AA}}{\sqrt{3}} = 1.4203 \times 10^{-10} \text{ m},$$

with

$$|t| = 2.70 \text{ eV}$$

and

$$\hbar = 6.582119569 \times 10^{-16} \text{ eV s},$$

we find

$$v_F = \frac{3(2.70)(1.4203 \times 10^{-10})}{2(6.582119569 \times 10^{-16})} \approx 8.74 \times 10^5 \text{ m/s}.$$

The linear dependence

$$E_{\pm} \propto |\mathbf{q}|$$

shows that, close to the  $K$  point, the conduction and valence bands have a linear dispersion rather than a quadratic one. This produces the Dirac-cone shape seen near the Dirac point. The parameter  $v_F$  is the slope of this linear dispersion and gives the characteristic velocity of electrons near  $K$ .

## 1.6 Question 6

Each primitive cell contains two carbon atoms, one on sublattice  $A$  and one on sublattice  $B$ . Each carbon contributes one relevant  $p_z$  orbital, so there are two basis orbitals per primitive cell.

Therefore, the Bloch Hamiltonian is a  $2 \times 2$  matrix and has two eigenvalues for each  $\mathbf{k}$ . These two eigenvalues form the two energy bands,

$$E_-(\mathbf{k}) \quad \text{and} \quad E_+(\mathbf{k}).$$

Each carbon also contributes one electron to the  $p_z$  system, so there are two  $p_z$  electrons per primitive cell.

Including spin, the two bands together can hold four electrons per primitive cell. Therefore, two electrons correspond to half filling.

At half filling, the lower band is completely occupied while the upper band is empty. In the nearest-neighbor graphene model, the two bands touch at the  $K$  and  $K'$  points at

$$E = 0.$$

Thus, the highest occupied states and the lowest unoccupied states meet at the same energy. The Fermi energy is therefore

$$E_F = 0.$$

Hence, half filling places the Fermi energy exactly at the band-touching points.

## 2 Problem 2

### 2.1 Question 1

Using the basis

$$(A_1, B_1, A_2, B_2),$$

the nearest-neighbor hoppings are

$$A_1 \leftrightarrow B_1$$

and

$$A_2 \leftrightarrow B_2,$$

with matrix elements

$$tf(\mathbf{k})$$

and their complex conjugates.  
The dimer hopping is between

$$B_1 \leftrightarrow A_2,$$

with coupling strength

$$\gamma_1.$$

Therefore, the Hamiltonian is

$$H_{2L}(\mathbf{k}) = \begin{pmatrix} 0 & tf(\mathbf{k}) & 0 & 0 \\ tf^*(\mathbf{k}) & 0 & \gamma_1 & 0 \\ 0 & \gamma_1 & 0 & tf(\mathbf{k}) \\ 0 & 0 & tf^*(\mathbf{k}) & 0 \end{pmatrix}.$$

The nonzero matrix elements correspond to the couplings

$$A_1 \leftrightarrow B_1, \quad B_1 \leftrightarrow A_2, \quad A_2 \leftrightarrow B_2.$$

The Hamiltonian is Hermitian because each hopping term has the corresponding reverse hopping as its complex conjugate.

## 2.2 Question 2

We diagonalize the Hamiltonian

$$H_{2L}(\mathbf{k}) = \begin{pmatrix} 0 & tf(\mathbf{k}) & 0 & 0 \\ tf^*(\mathbf{k}) & 0 & \gamma_1 & 0 \\ 0 & \gamma_1 & 0 & tf(\mathbf{k}) \\ 0 & 0 & tf^*(\mathbf{k}) & 0 \end{pmatrix}.$$

The energy eigenvalues satisfy

$$\det(H_{2L}(\mathbf{k}) - EI) = 0.$$

For convenience, define

$$\alpha = |tf(\mathbf{k})|^2.$$

The characteristic equation becomes

$$E^4 - (2\alpha + \gamma_1^2) E^2 + \alpha^2 = 0.$$

Let

$$x = E^2.$$

Then

$$x^2 - (2\alpha + \gamma_1^2)x + \alpha^2 = 0.$$

Using the quadratic formula,

$$x = \frac{2\alpha + \gamma_1^2 \pm \sqrt{(2\alpha + \gamma_1^2)^2 - 4\alpha^2}}{2}.$$

The expression under the square root simplifies to

$$\gamma_1^4 + 4\gamma_1^2\alpha,$$

so

$$E^2 = \alpha + \frac{\gamma_1^2}{2} \pm \frac{\gamma_1}{2} \sqrt{\gamma_1^2 + 4\alpha}.$$

This can be written in the form

$$E_{s,\lambda}(\mathbf{k}) = s \left[ \sqrt{|tf(\mathbf{k})|^2 + \left(\frac{\gamma_1}{2}\right)^2} + \lambda \frac{\gamma_1}{2} \right],$$

where

$$s = \pm 1, \quad \lambda = \pm 1.$$

The four sign combinations give the four energy bands. The sign  $s$  determines whether the energy is positive or negative, while  $\lambda$  is responsible for the branches split by the coupling  $\gamma_1$ .

## 2.3 Question 3

The four bands were plotted along the high-symmetry path

$$\Gamma \rightarrow K \rightarrow M \rightarrow \Gamma$$

using the energies obtained in Question 2. The result was compared with two uncoupled graphene layers, corresponding to

$$\gamma_1 = 0.$$

For the uncoupled case, each monolayer band is doubly degenerate because the two layers are independent. Thus, the two positive-energy bands coincide and the two negative-energy bands also coincide.

When the interlayer coupling

$$\gamma_1 = 0.390 \text{ eV}$$

is included, this degeneracy is lifted and the four bilayer bands separate.

The effect is clearest at the  $K$  point. For two uncoupled layers, all four bands meet at

$$E = 0.$$

For the AB-stacked bilayer, the vertical dimer coupling between  $B_1$  and  $A_2$  shifts two of these states away from zero, giving

$$E = -\gamma_1, \quad 0, \quad 0, \quad +\gamma_1.$$

Therefore, at  $K$ ,

$$E = -0.390 \text{ eV}, \quad 0, \quad 0, \quad +0.390 \text{ eV}.$$

The two low-energy bands still touch at zero energy, while the other two bands are split away from the touching point by the interlayer coupling.

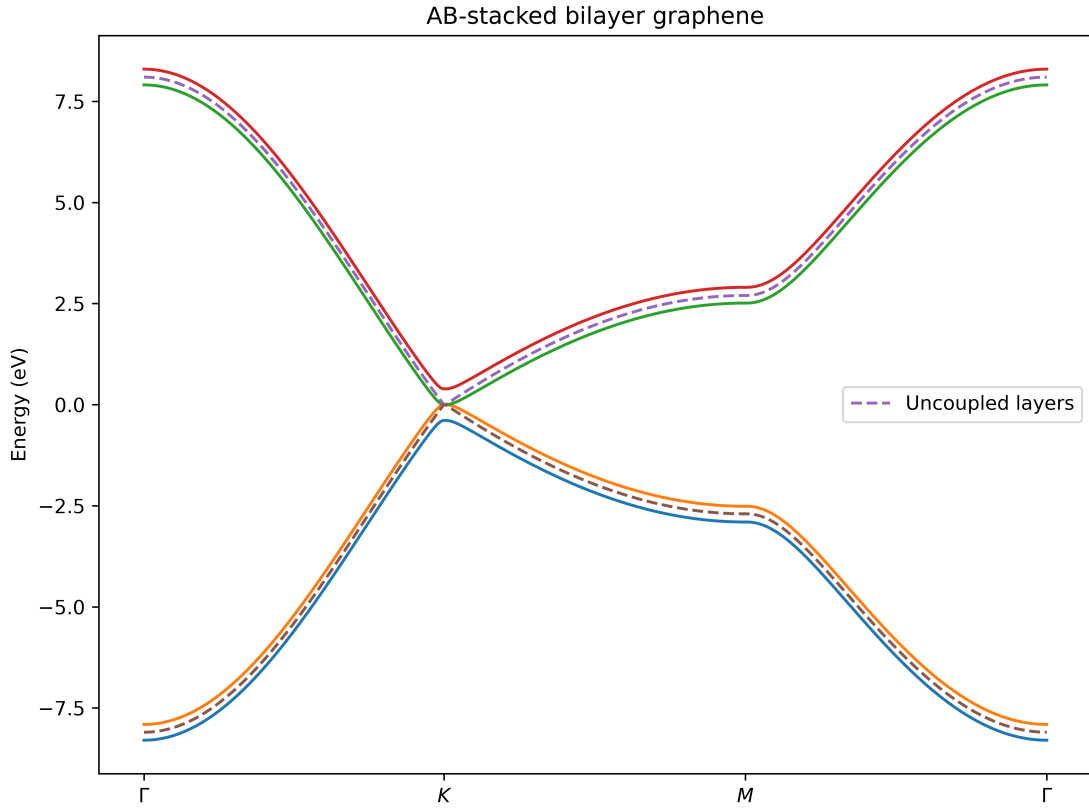


Figure 2: Band structure of AB-stacked bilayer graphene along  $\Gamma \rightarrow K \rightarrow M \rightarrow \Gamma$ . The solid curves show the four coupled bilayer bands, while the dashed curves show the doubly degenerate bands of two uncoupled graphene layers.

## 2.4 Question 4

At the  $K$  point, the Hamiltonian eigenvalues are

$$E = -0.390 \text{ eV}, \quad E = 0, \quad E = 0, \quad E = 0.390 \text{ eV}.$$

The corresponding orbital weights are

| $E \text{ (eV)}$ | $ c_{A_1} ^2$ | $ c_{B_1} ^2$ | $ c_{A_2} ^2$ | $ c_{B_2} ^2$ |
|------------------|---------------|---------------|---------------|---------------|
| -0.390           | 0.00          | 0.50          | 0.50          | 0.00          |
| 0.000            | 1.00          | 0.00          | 0.00          | 0.00          |
| 0.000            | 0.00          | 0.00          | 0.00          | 1.00          |
| 0.390            | 0.00          | 0.50          | 0.50          | 0.00          |

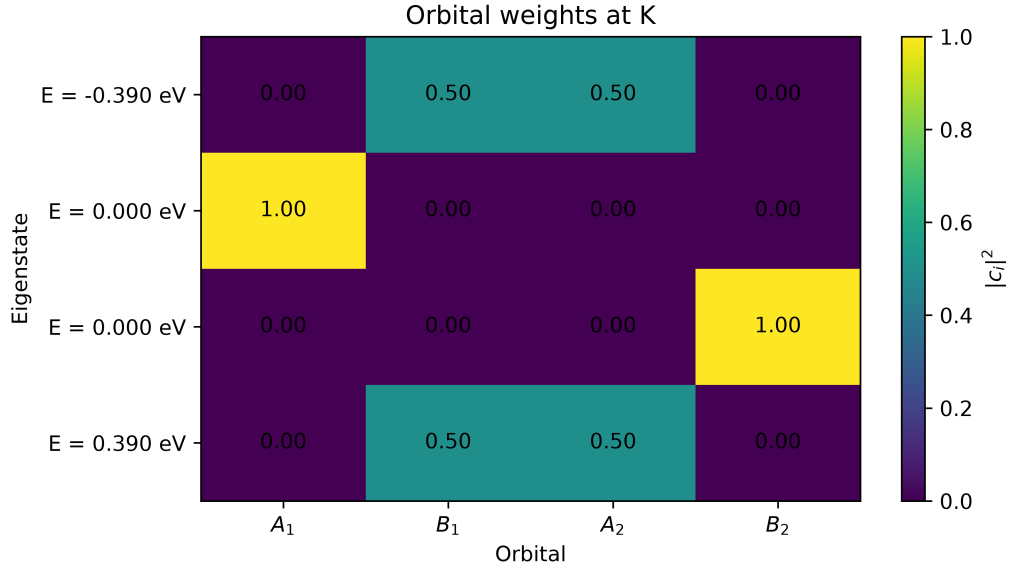


Figure 3: Orbital weights of the four bilayer eigenstates at the  $K$  point. The states at  $E = \pm\gamma_1$  are supported on the dimer orbitals  $B_1$  and  $A_2$ , while the two zero-energy states are supported on the non-dimer orbitals  $A_1$  and  $B_2$ .

## 2.5 Question 5

At the  $K$  point,

$$f(K) = 0,$$

so the intralayer hopping terms vanish.

The only remaining coupling is the interlayer coupling between the dimer sites

$$B_1 \leftrightarrow A_2$$

with strength  $\gamma_1$ .

The eigenstates at

$$E = \pm\gamma_1$$

therefore have equal orbital weight on  $B_1$  and  $A_2$ ,

$$|c_{B_1}|^2 = |c_{A_2}|^2 = 0.50.$$

Thus,  $B_1$  and  $A_2$  are the dimer sites.

The two zero-energy eigenstates instead have weight on

$$A_1$$

and

$$B_2,$$

which are the non-dimer sites.

Because these two sites are not directly coupled by the remaining  $K$ -point Hamiltonian, their states remain at

$$E = 0.$$

Therefore,

$$B_1, A_2$$

form the dimer pair and produce the split energies

$$E = \pm\gamma_1,$$

while

$$A_1, B_2$$

are the non-dimer sites and produce the two zero-energy states.

## 2.6 Question 6

In the single-layer problem, the primitive cell contains two orbitals,

$$A, B.$$

Therefore, the Bloch basis contains two states and the Hamiltonian is a  $2 \times 2$  matrix. Diagonalizing this matrix gives two eigenvalues at each  $\mathbf{k}$ , producing two energy bands.

For the AB-stacked bilayer, the repeat unit contains four orbitals,

$$A_1, B_1, A_2, B_2.$$

The Bloch basis therefore contains four states, so the Hamiltonian becomes a  $4 \times 4$  matrix. Diagonalizing this matrix gives four eigenvalues at each  $\mathbf{k}$ , producing four energy bands.

Thus, increasing the number of orbitals in the repeat unit increases the number of bands,

$$2 \text{ orbitals} \rightarrow 2 \text{ bands},$$

4 orbitals  $\rightarrow$  4 bands.

The basic Bloch construction does not change. In both cases, Bloch sums are formed for the orbitals in the repeat unit, the Hamiltonian is written in that Bloch basis, and the matrix is diagonalized for each  $\mathbf{k}$ . Only the size of the basis and therefore the size of the Hamiltonian change.

### 3 Problem 3

#### 3.1 Question 1

For bulk Bernal graphite, the AB-stacked structure repeats periodically along the  $z$  direction. We use the same four-orbital basis as in the bilayer problem,

$$(A_1, B_1, A_2, B_2).$$

The in-plane nearest-neighbor hoppings remain

$$A_1 \leftrightarrow B_1$$

and

$$A_2 \leftrightarrow B_2,$$

with matrix elements

$$tf(k_x, k_y)$$

and their complex conjugates.

The vertical dimer coupling is modified because the graphite structure repeats above and below along  $z$ . The two vertical hopping contributions acquire Bloch phases

$$\gamma_1 e^{ik_z c/2}$$

and

$$\gamma_1 e^{-ik_z c/2}.$$

These two amplitudes add to give

$$g(k_z) = \gamma_1 e^{ik_z c/2} + \gamma_1 e^{-ik_z c/2}.$$

Using

$$e^{ix} + e^{-ix} = 2 \cos x,$$

we obtain

$$g(k_z) = 2\gamma_1 \cos\left(\frac{k_z c}{2}\right).$$

Therefore, the bulk Bloch Hamiltonian is

$$H(\mathbf{k}) = \begin{pmatrix} 0 & tf(k_x, k_y) & 0 & 0 \\ tf^*(k_x, k_y) & 0 & g(k_z) & 0 \\ 0 & g(k_z) & 0 & tf(k_x, k_y) \\ 0 & 0 & tf^*(k_x, k_y) & 0 \end{pmatrix}.$$

The two physically distinct vertical hoppings correspond to hopping to the above repeat and to the below repeat. In the Bloch basis, these two contributions combine into the single matrix element

$$g(k_z) = 2\gamma_1 \cos\left(\frac{k_z c}{2}\right).$$

### 3.2 Question 2

The four bulk graphite bands were calculated along the three-dimensional high-symmetry path

$$\Gamma \rightarrow K \rightarrow M \rightarrow \Gamma \rightarrow A \rightarrow H \rightarrow L \rightarrow A.$$

The Hamiltonian was diagonalized at each wavevector along the path using

$$g(k_z) = 2\gamma_1 \cos\left(\frac{k_z c}{2}\right).$$

The resulting band structure shows a clear dependence on  $k_z$ . The bands are split in the  $k_z = 0$  plane, while the splitting decreases as the path moves toward the  $k_z = \pi/c$  plane.

At the  $H$  point, all four bands meet at zero energy in this minimal model.

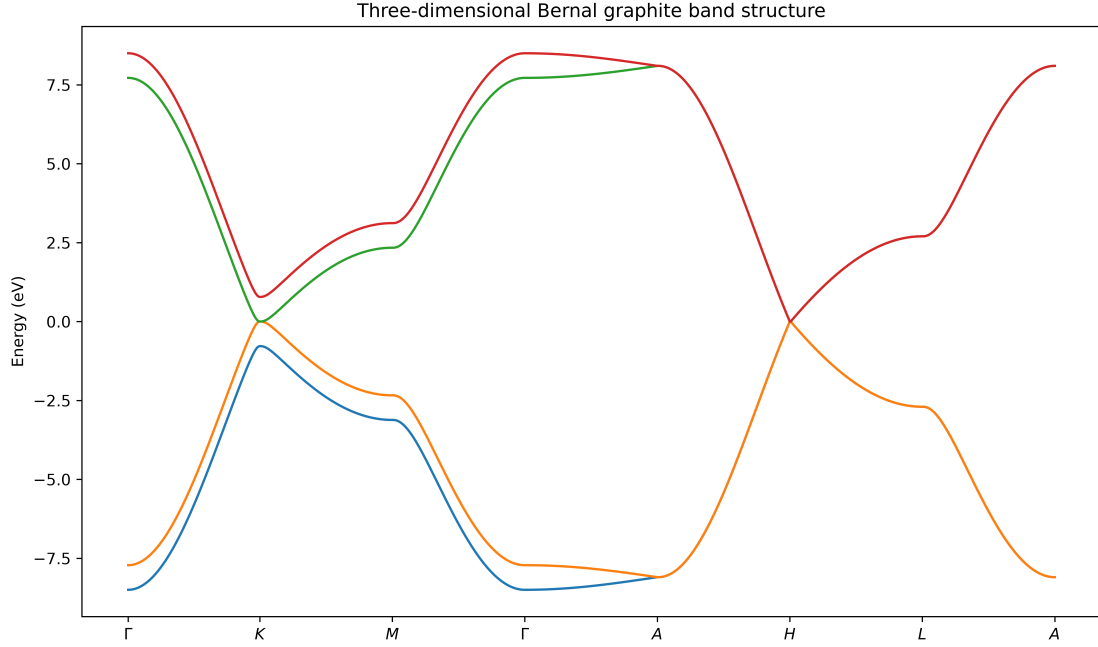


Figure 4: Three-dimensional Bernal graphite band structure along  $\Gamma \rightarrow K \rightarrow M \rightarrow \Gamma \rightarrow A \rightarrow H \rightarrow L \rightarrow A$ .

### 3.3 Question 3

At the  $K$  point,

$$f(K) = 0$$

and

$$k_z = 0.$$

Therefore,

$$g(0) = 2\gamma_1 \cos(0) = 2\gamma_1.$$

The Hamiltonian at  $K$  reduces to

$$H(K) = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 2\gamma_1 & 0 \\ 0 & 2\gamma_1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Its four eigenvalues are

$$E = -2\gamma_1, \quad 0, \quad 0, \quad +2\gamma_1.$$

Using

$$\gamma_1 = 0.390 \text{ eV},$$

this gives

$$E = -0.780 \text{ eV}, \quad 0, \quad 0, \quad +0.780 \text{ eV}.$$

At the  $H$  point, the in-plane coordinates are the same as at  $K$ , so

$$f(H) = 0,$$

but now

$$k_z = \frac{\pi}{c}.$$

Therefore,

$$g\left(\frac{\pi}{c}\right) = 2\gamma_1 \cos\left(\frac{\pi}{2}\right) = 0.$$

Thus, in this minimal model,

$$H(H) = 0,$$

and all four energies are

$$E = 0, \quad 0, \quad 0, \quad 0.$$

The difference between  $K$  and  $H$  comes from the vertical Bloch phases. At

$$k_z = 0,$$

the two vertical hopping contributions are in phase,

$$e^{ik_z c/2} + e^{-ik_z c/2} = 1 + 1 = 2,$$

so the effective interlayer coupling is maximal,

$$g(0) = 2\gamma_1.$$

At

$$k_z = \frac{\pi}{c},$$

the two contributions have opposite phases,

$$e^{i\pi/2} + e^{-i\pi/2} = i - i = 0,$$

so they cancel and the interlayer coupling vanishes.

Therefore, the layers are maximally coupled at  $K$  and decouple at  $H$  in this minimal model.

### 3.4 Question 4

The full-zone density of states was calculated numerically using a shifted uniform mesh spanning one primitive reciprocal cell.

A mesh of

$$24 \times 24 \times 24$$

$k$ -points was used, corresponding to

$$N_k = 13824$$

wavevectors in the primitive reciprocal cell.

At each wavevector, the full  $4 \times 4$  bulk Hamiltonian was diagonalized to obtain the four band energies.

The density of states was constructed using Gaussian broadening,

$$D(E) = \frac{1}{N_k} \sum_{\mathbf{k}, n} \frac{1}{\eta \sqrt{\pi}} \exp \left[ - \left( \frac{E - E_{n\mathbf{k}}}{\eta} \right)^2 \right],$$

with broadening

$$\eta = 0.08 \text{ eV}.$$

The DOS was normalized by the total number of sampled wavevectors, so that it gives the number of states per unit energy per primitive cell.

The numerical integration gives

$$\int D(E) dE = 3.999999999999993 \approx 4.$$

This agrees with the expected four states per primitive cell when spin is not counted, since the primitive cell contains four  $p_z$  orbitals and therefore four bands.

The calculated DOS is approximately symmetric around

$$E = 0,$$

as expected for this minimal model. The DOS becomes very small near the Fermi energy, consistent with the band-touching structure of the model. Large DOS features occur at energies where many  $k$ -states have similar band energies.

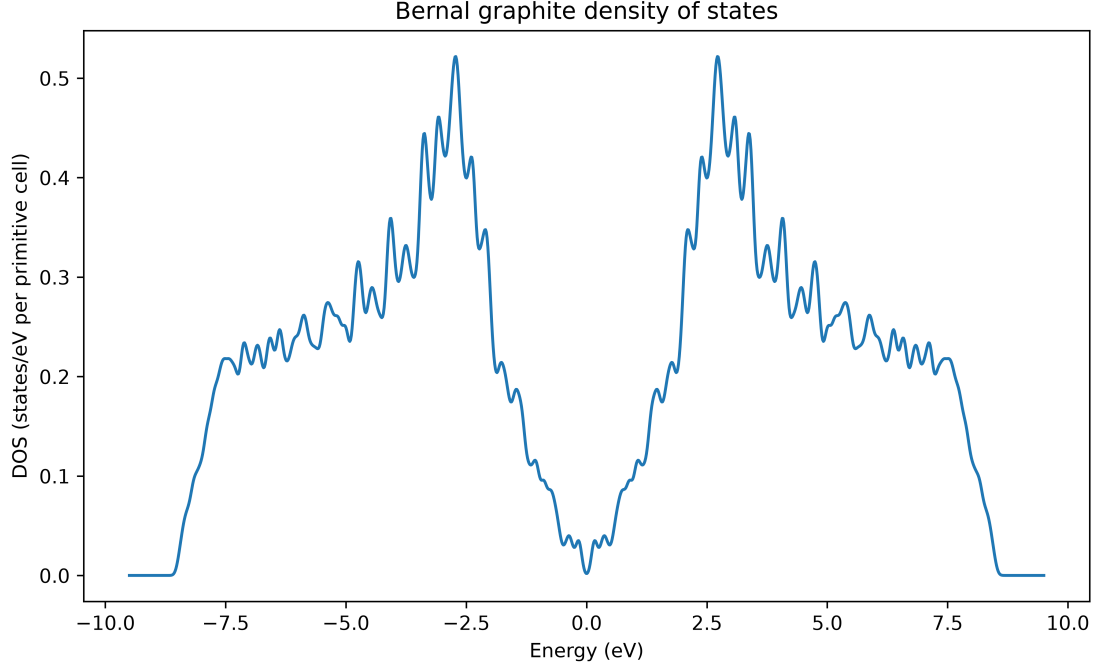


Figure 5: Density of states for the minimal Bernal graphite model calculated on a shifted  $24 \times 24 \times 24$  reciprocal-space mesh using a Gaussian broadening of 0.08 eV.

### 3.5 Question 5

Harrison's rule gives the vertical  $p_z$ - $p_z$  sigma hopping as

$$V_{pp\sigma} = 2.22 \frac{\hbar^2}{m_e d_{\perp}^2}.$$

Using

$$\frac{\hbar^2}{m_e} = 7.62 \text{ eV } \text{\AA}^2$$

and

$$d_{\perp} = 3.35 \text{ \AA},$$

we obtain

$$V_{pp\sigma} = 2.22 \frac{7.62}{(3.35)^2} \text{ eV}.$$

Since

$$(3.35)^2 = 11.2225,$$

this gives

$$V_{pp\sigma} \approx 2.22 \left( \frac{7.62}{11.2225} \right) \approx 1.51 \text{ eV}.$$

The empirical graphite interlayer hopping used in the tight-binding model is

$$\gamma_1 = 0.390 \text{ eV}.$$

Therefore,

$$\frac{V_{pp\sigma}}{\gamma_1} \approx \frac{1.51}{0.390} \approx 3.9.$$

Thus, Harrison's estimate is approximately four times larger than the empirical graphite value.

Harrison's rule provides a simple two-center orbital-overlap estimate. In real graphite, the relatively large separation between layers produces a weak interlayer bond, and the effective hopping is influenced by the detailed orbital environment of the crystal. Therefore, the empirical value

$$\gamma_1 = 0.390 \text{ eV}$$

is more appropriate for describing the actual interlayer coupling in graphite.

### 3.6 Question 6

In this minimal model, the Fermi energy is at

$$E_F = 0.$$

The bands touch at the Fermi energy, so the model describes graphite as a zero-gap semimetal.

For a more accurate graphite Fermi surface, additional effects must be included. Examples are the hopping terms

$$\gamma_2$$

and

$$\gamma_3.$$

These extra hopping terms modify the low-energy bands and make the model more realistic.

## 4 Python

Documented files for each problem are submitted separately.